Tsunami Risk Assessment Report

generated using RAFT-Ports by Felix, et al. (2026)

Contents:

Inundated Structures
Structural Damage States
Physical Loss Ratio (Repair / Replacement)
Recovery of Production Capacity

Input Files:

Inundation: Mw89_FlowDepth.tif
File Resolution: 4.4870e-05, 4.4870e-05 degree
Structures: structures.shp

Raster Resampling Method: Nearest
Resampling Factor: 2.00
Updated Resolution: 2.243516e-05, 2.243486e-05 degree
Depth extracted within the building mask: Maximum
Probability Threshold: 0.70
Only structures meeting or exceeding this threshold are counted in the results

Report created by: Raquel Felix
Date created: 14-Aug-2026

RAFT-Ports: Risk Assessment for Tsunamis in Ports. RAFT-Ports provides scenario-based tsunami impact assessments to support risk-informed decision-making. Results are subject to uncertainties in input data, model assumptions, and the fragility and vulnerability functions used in the analysis. Users should independently review and validate all inputs and results prior to application.

Overview of the Results

Based on the analysis of 430 structures

Inundated Structures

total	chemical	petrochemical
418	334	84

Structural Damage States

minor	moderate	major	destroyed
416	0	0	0

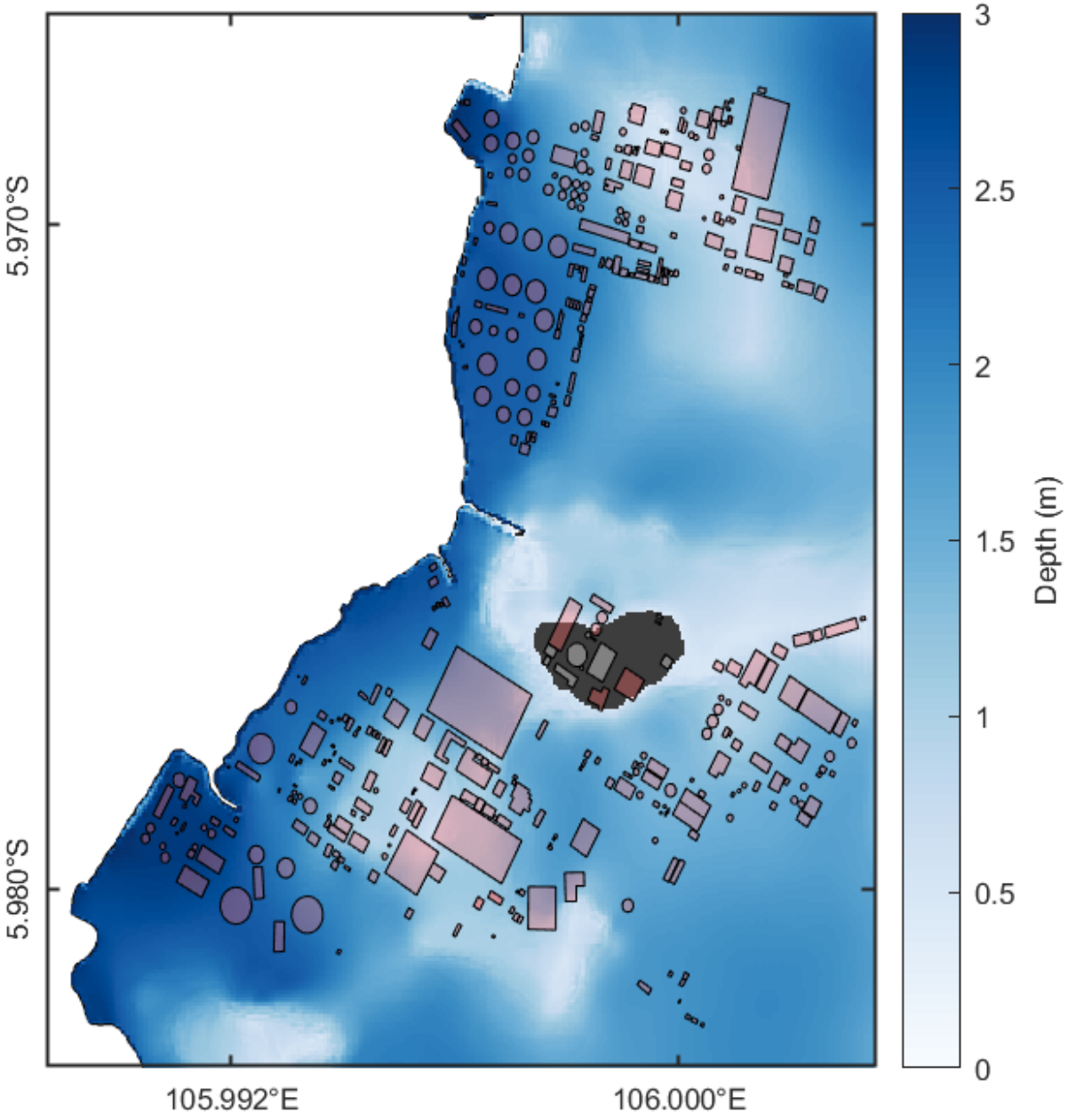
Physical Loss Ratio (Repair / Replacement)

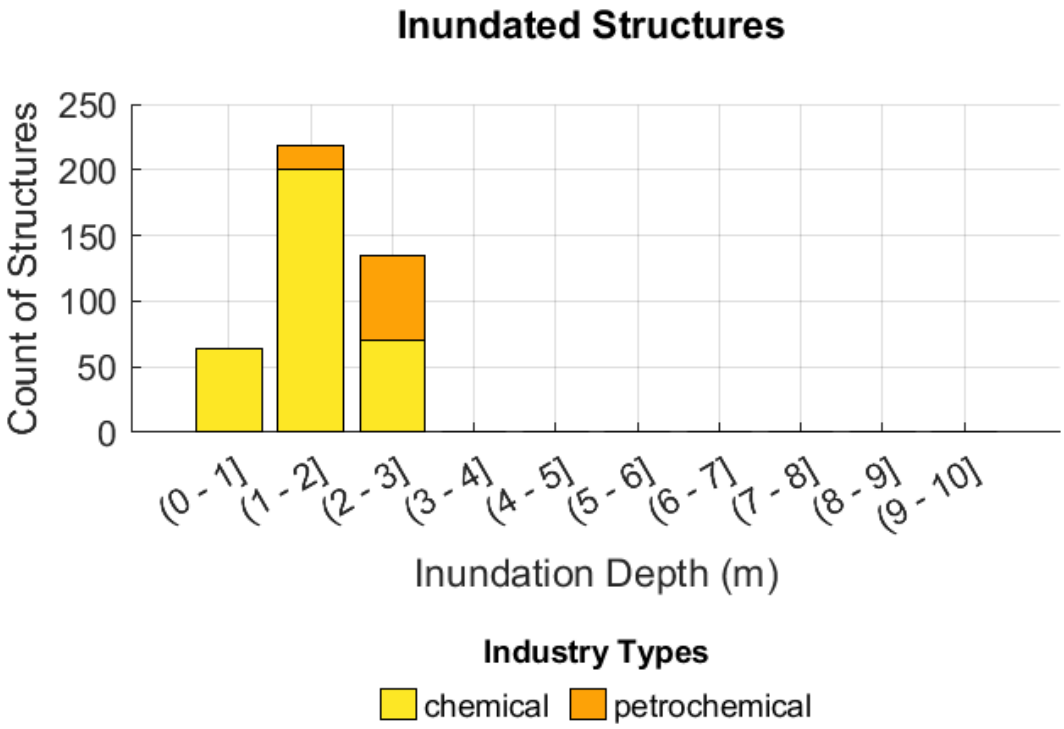
total	chemical	petrochemical
135	70	65

Potential Recovery Period of Production Capacity

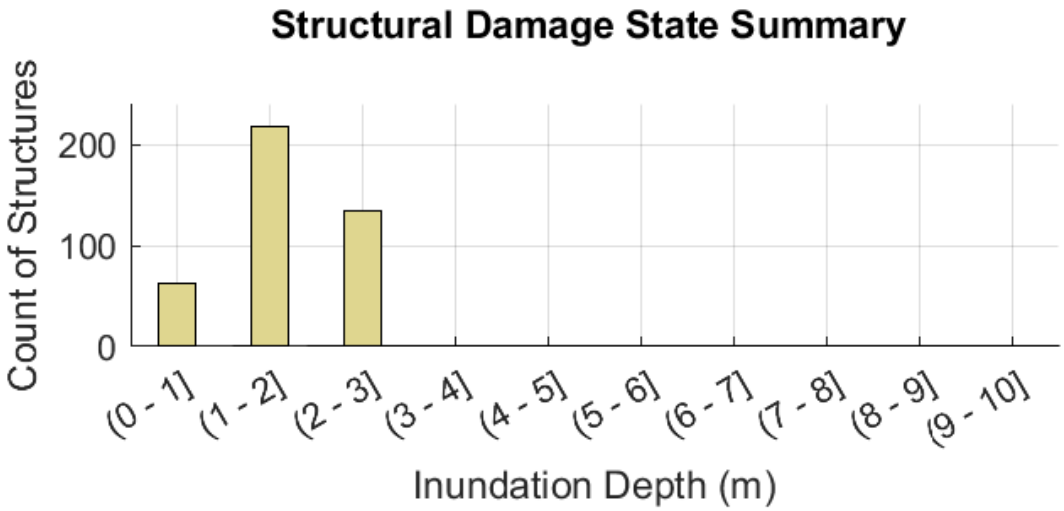
Potential Recovery	Month	Structure Count
Start of Recovery	(2.0 - 2.5]	12
Full Recovery	(5.0 - 5.5]	430

Structures Affected by Inundation





depth	total	chemical	petrochemical
(0 - 1]	64	64	0
(1 - 2]	219	200	19
(2 - 3]	135	70	65
(3 - 4]	0	0	0
(4 - 5]	0	0	0
(5 - 6]	0	0	0
(6 - 7]	0	0	0
(7 - 8]	0	0	0
(8 - 9]	0	0	0
(9 - 10]	0	0	0



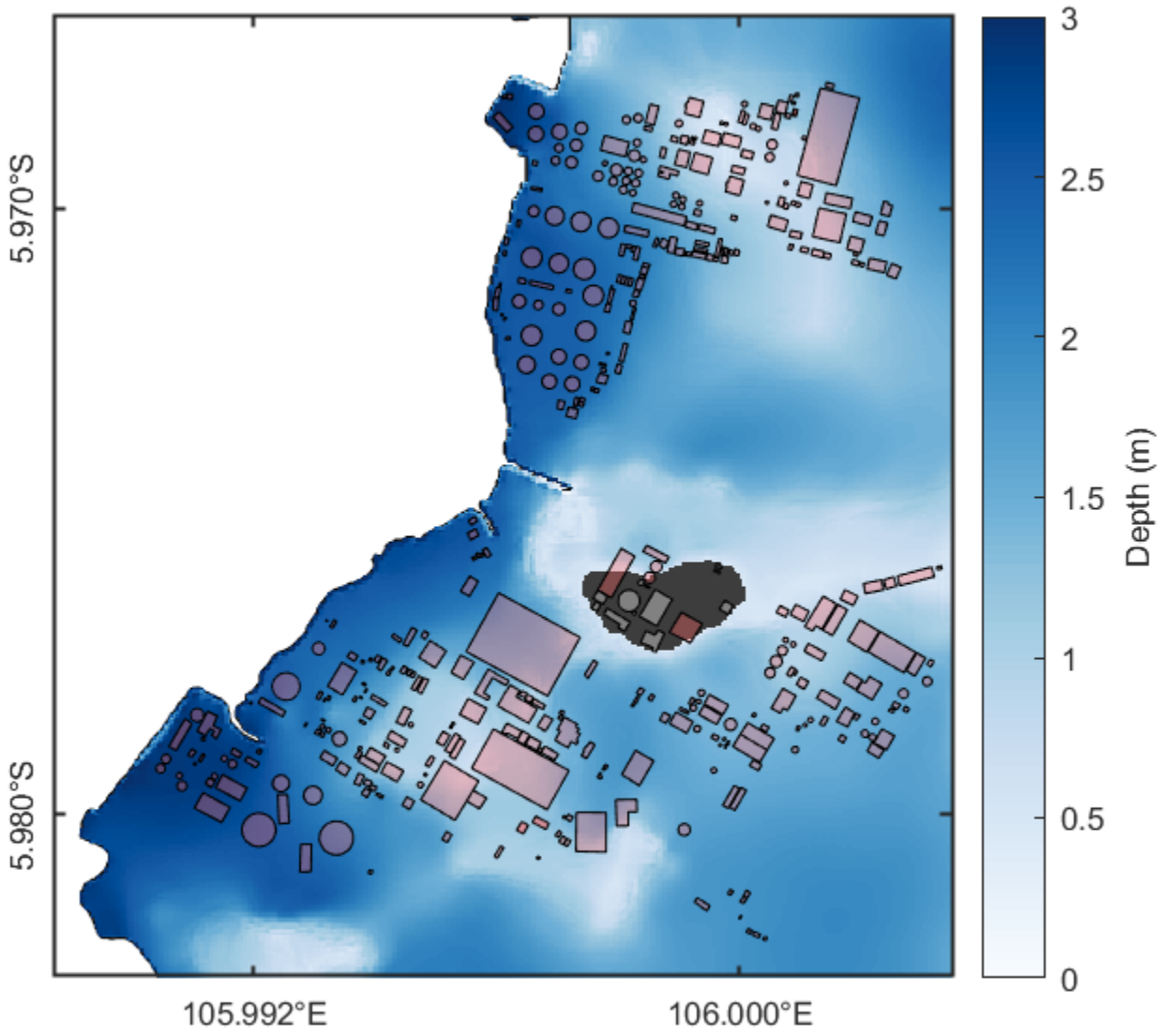
Count of Structures with Probability of Exceedance ≥ 0.70

Damage State

minor moderate major destroyed

depth	minor	moderate	major	destroyed
(0 - 1]	62	0	0	0
(1 - 2]	219	0	0	0
(2 - 3]	135	0	0	0
(3 - 4]	0	0	0	0
(4 - 5]	0	0	0	0
(5 - 6]	0	0	0	0
(6 - 7]	0	0	0	0
(7 - 8]	0	0	0	0
(8 - 9]	0	0	0	0
(9 - 10]	0	0	0	0

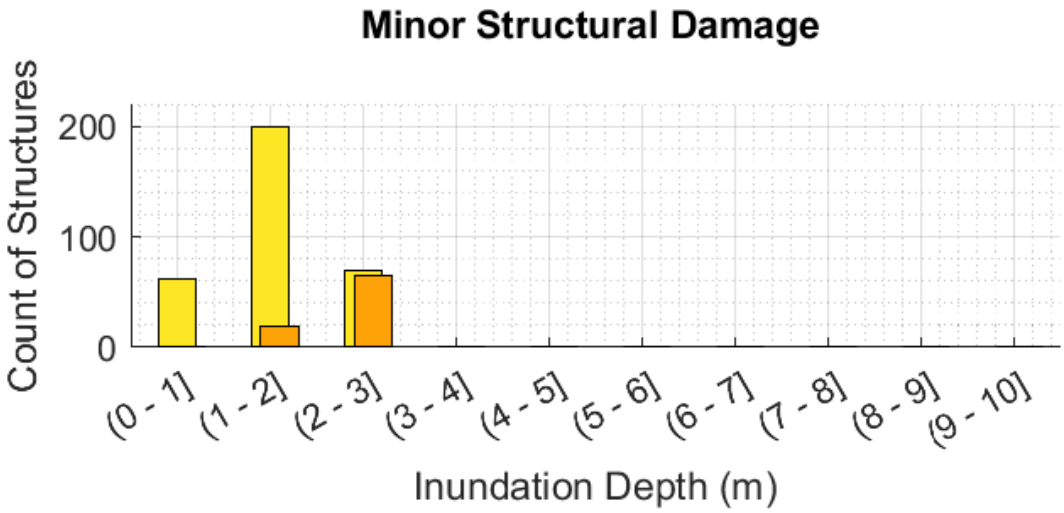
Minor Structural Damage



Structure Probability of Exceedance

■ ≥ 0.70 □ < 0.70

A total of 416 structures have a probability of exceedance ≥ 70% for the selected damage state.

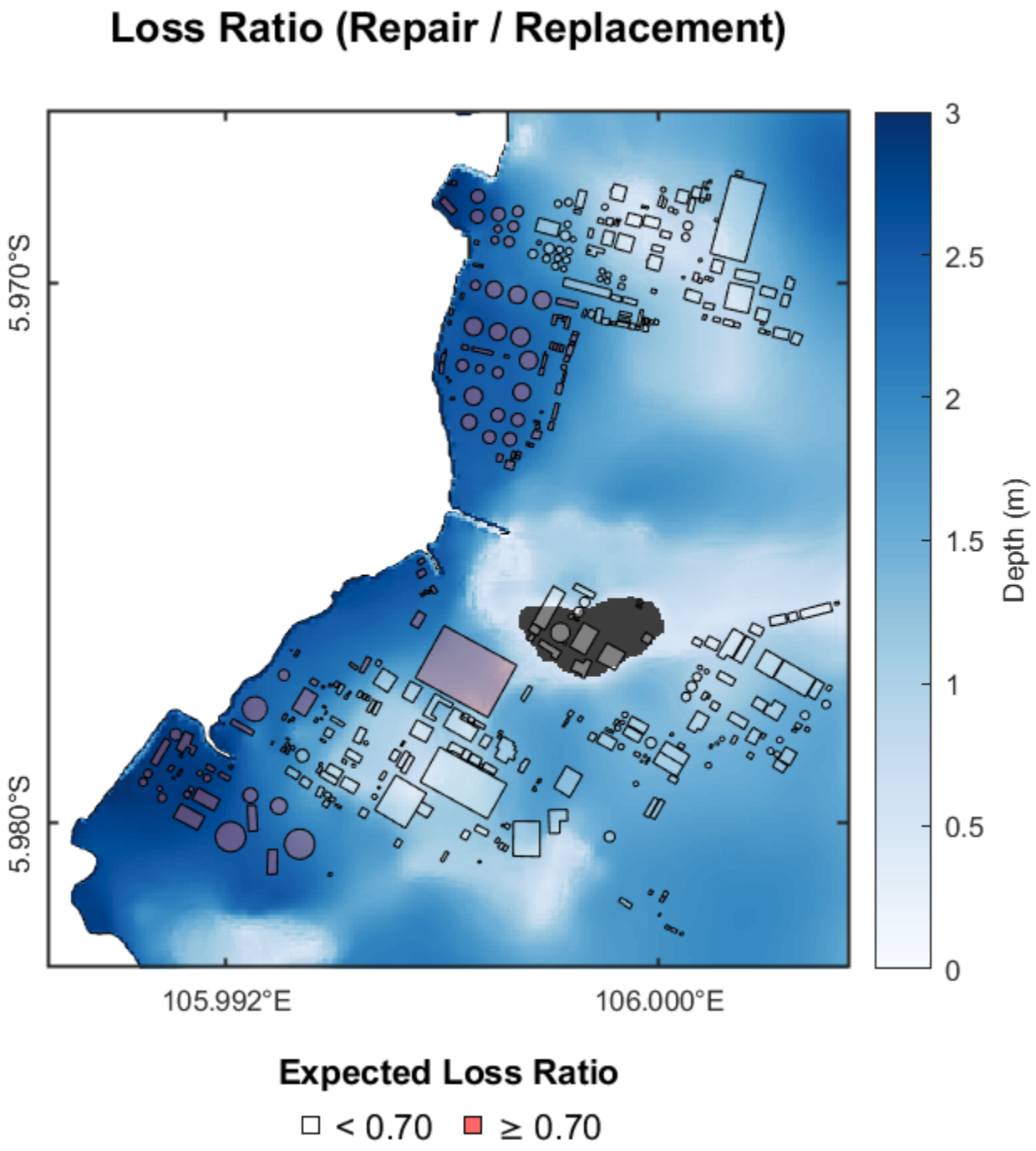


Count of Structures with Probability of Exceedance ≥ 0.70

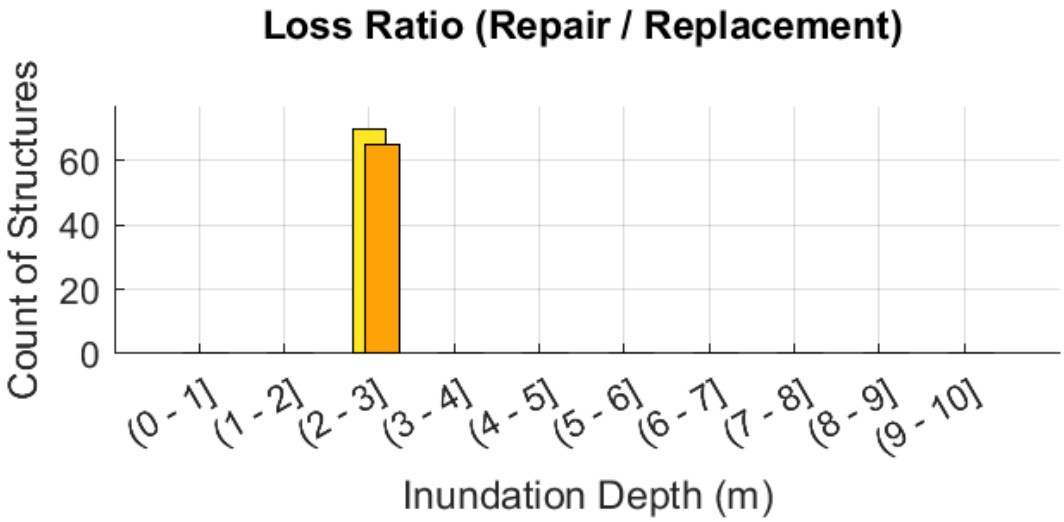
Industry Type

Chemical Petrochemical

depth	Total	chemical	petrochemical
(0 - 1]	62	62	0
(1 - 2]	219	200	19
(2 - 3]	135	70	65
(3 - 4]	0	0	0
(4 - 5]	0	0	0
(5 - 6]	0	0	0
(6 - 7]	0	0	0
(7 - 8]	0	0	0
(8 - 9]	0	0	0
(9 - 10]	0	0	0



A total of 135 structures have an expected loss ratio of ≥ 70%.



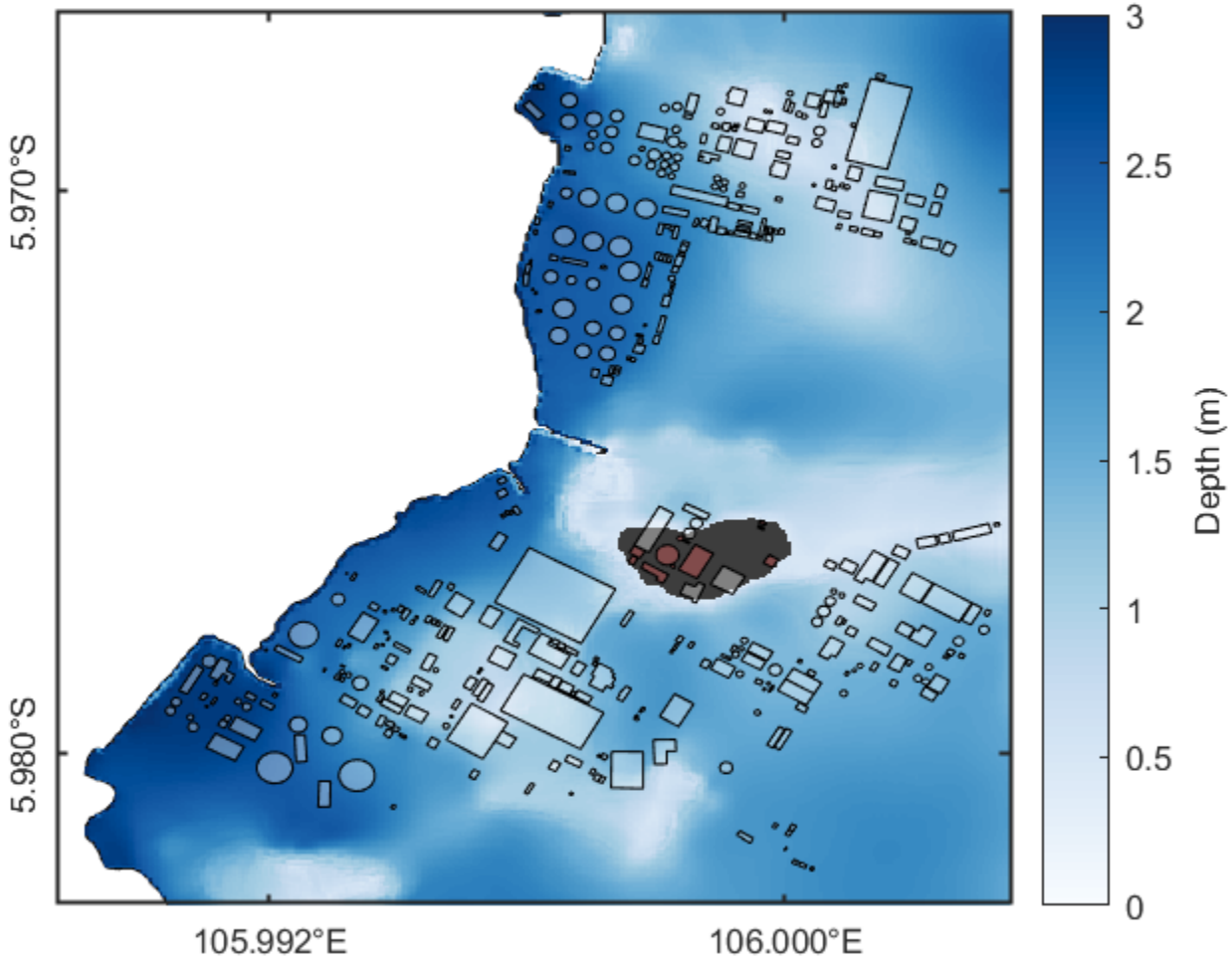
Count of structures with an expected loss ratio ≥ 0.70

Industry Type

chemical petrochemical

Depth	Total	chemical	petrochemical
(0 - 1]	0	0	0
(1 - 2]	0	0	0
(2 - 3]	135	70	65
(3 - 4]	0	0	0
(4 - 5]	0	0	0
(5 - 6]	0	0	0
(6 - 7]	0	0	0
(7 - 8]	0	0	0
(8 - 9]	0	0	0
(9 - 10]	0	0	0

Structures with a probability of recovery exceeding the threshold within 2.0 to ≤2.5 months

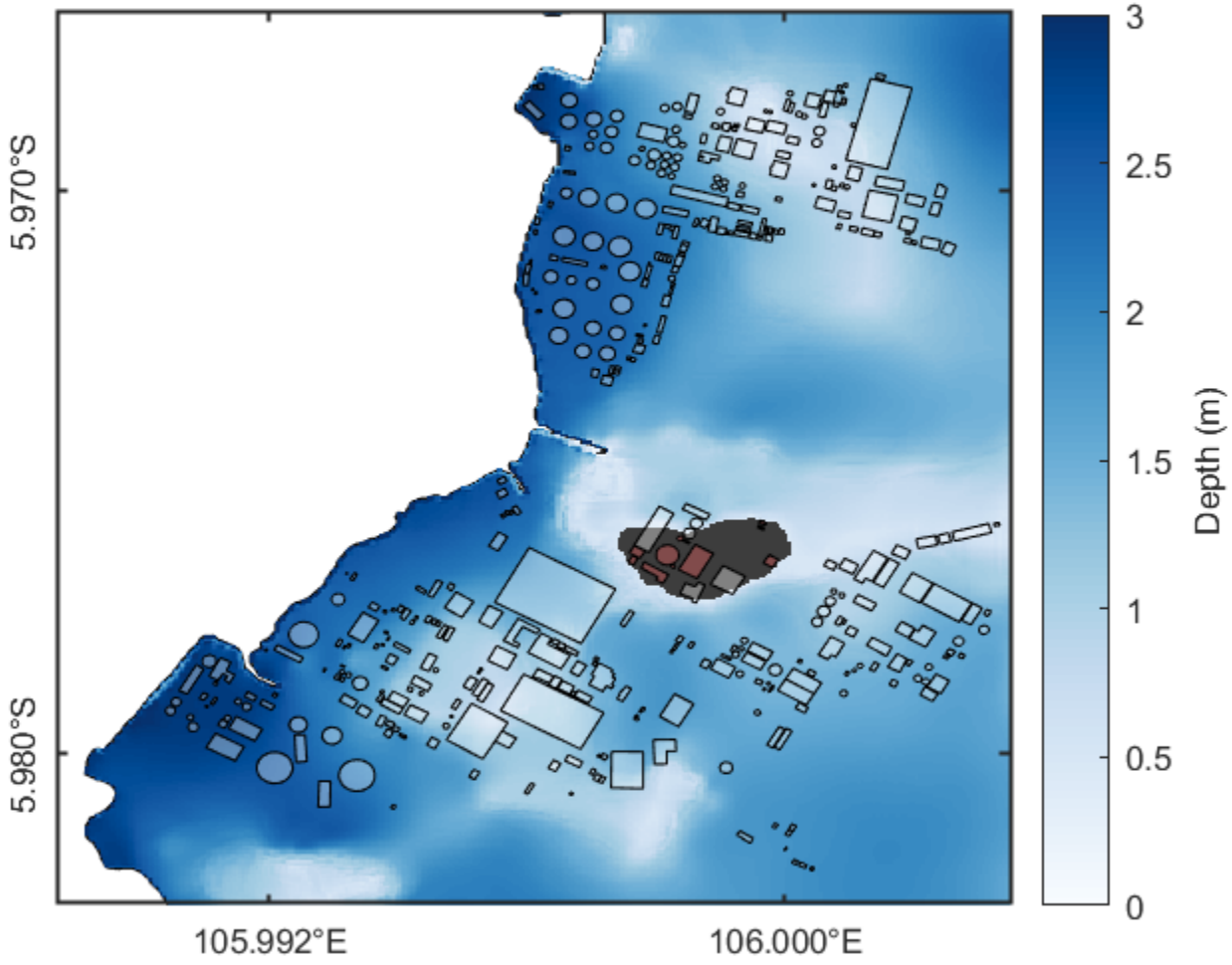


Structure Probability of Exceedance

□ < 0.70 ■ ≥ 0.70

A total of 12 structures have a probability of exceedance ≥ 70% for the selected recovery period.

Structures with a probability of recovery exceeding the threshold within 2.5 to ≤3.0 months

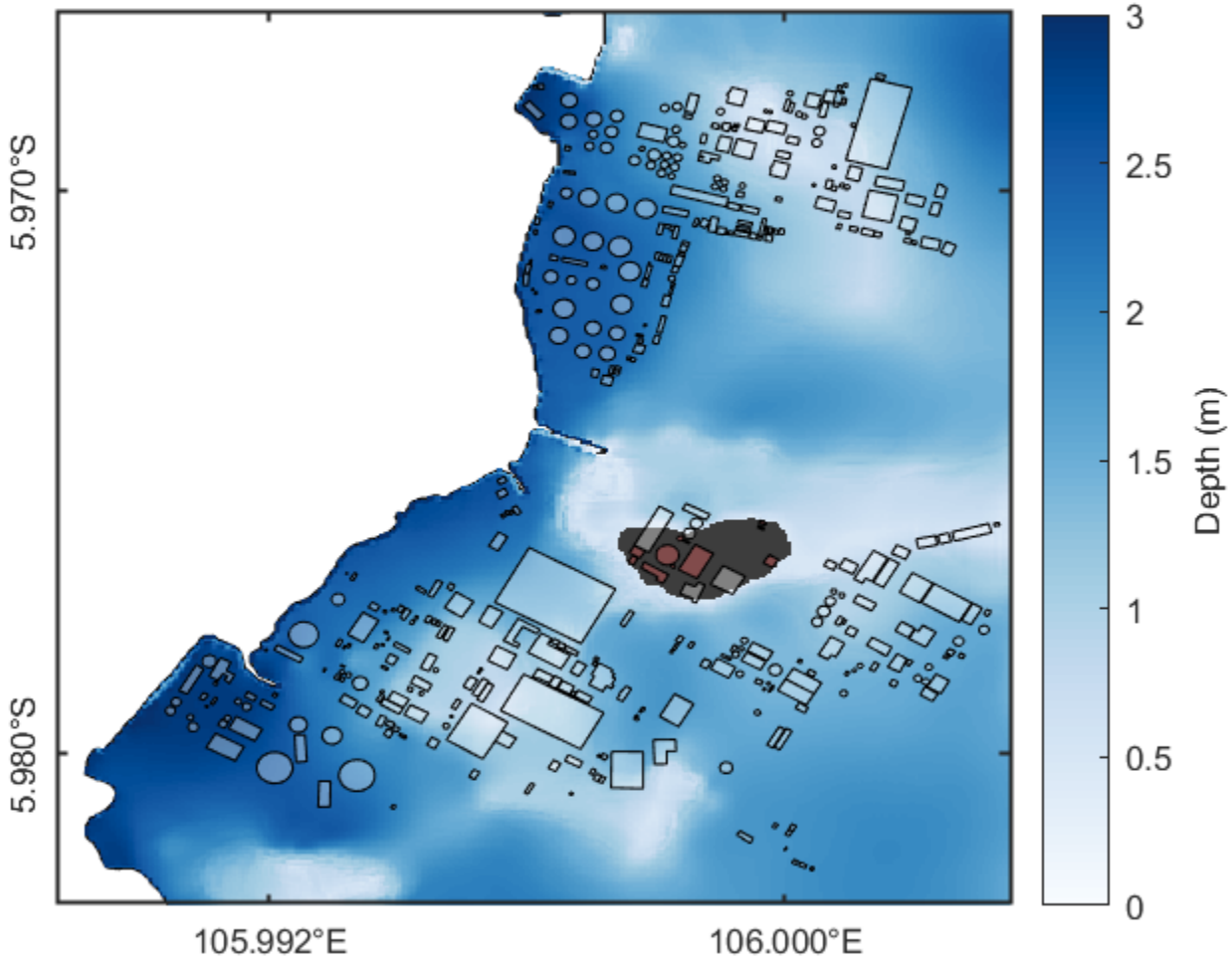


Structure Probability of Exceedance

□ < 0.70 ■ ≥ 0.70

A total of 12 structures have a probability of exceedance ≥ 70% for the selected recovery period.

Structures with a probability of recovery exceeding the threshold within 3.0 to ≤3.5 months

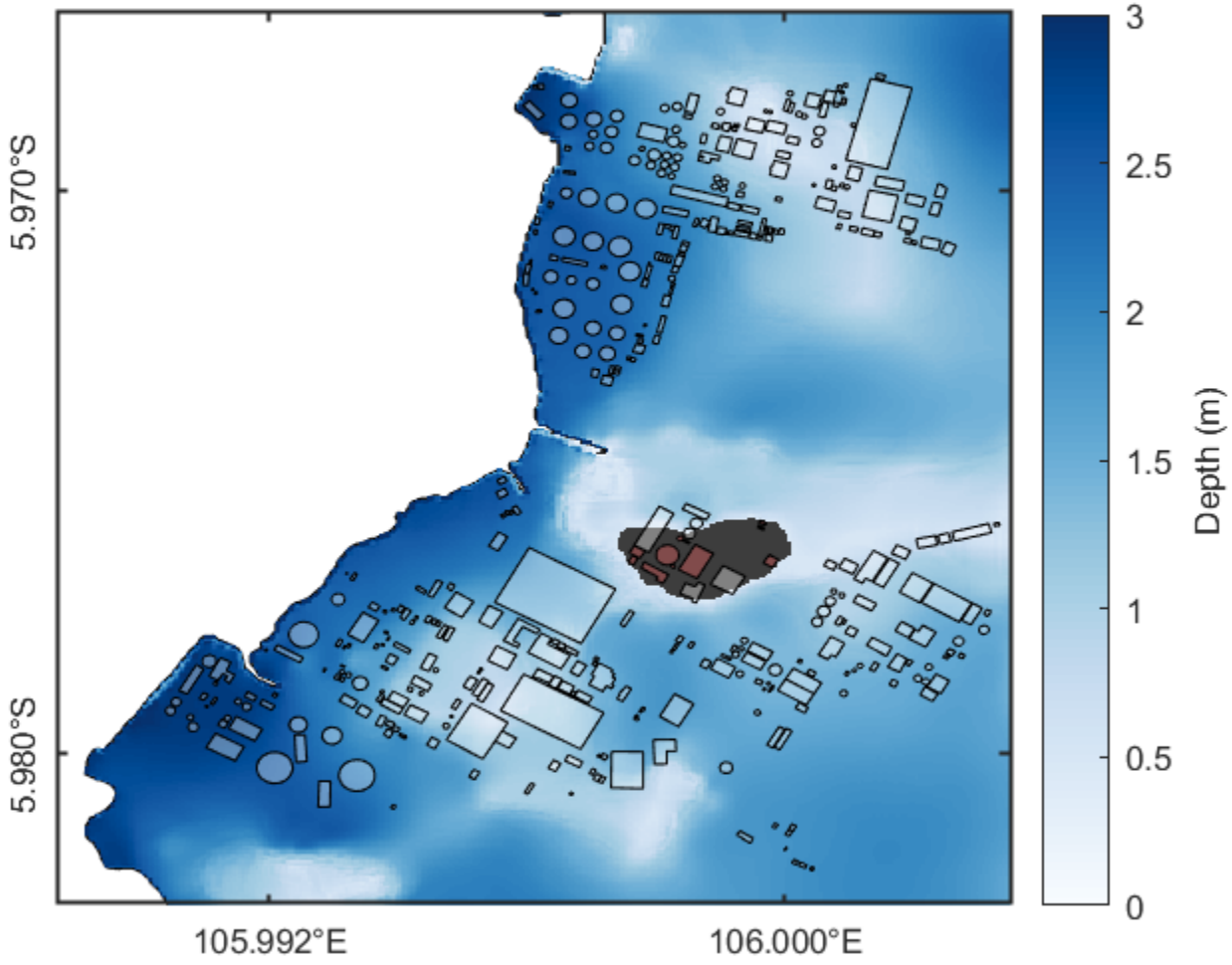


Structure Probability of Exceedance

□ < 0.70 ■ ≥ 0.70

A total of 12 structures have a probability of exceedance ≥ 70% for the selected recovery period.

Structures with a probability of recovery exceeding the threshold within 3.5 to ≤4.0 months

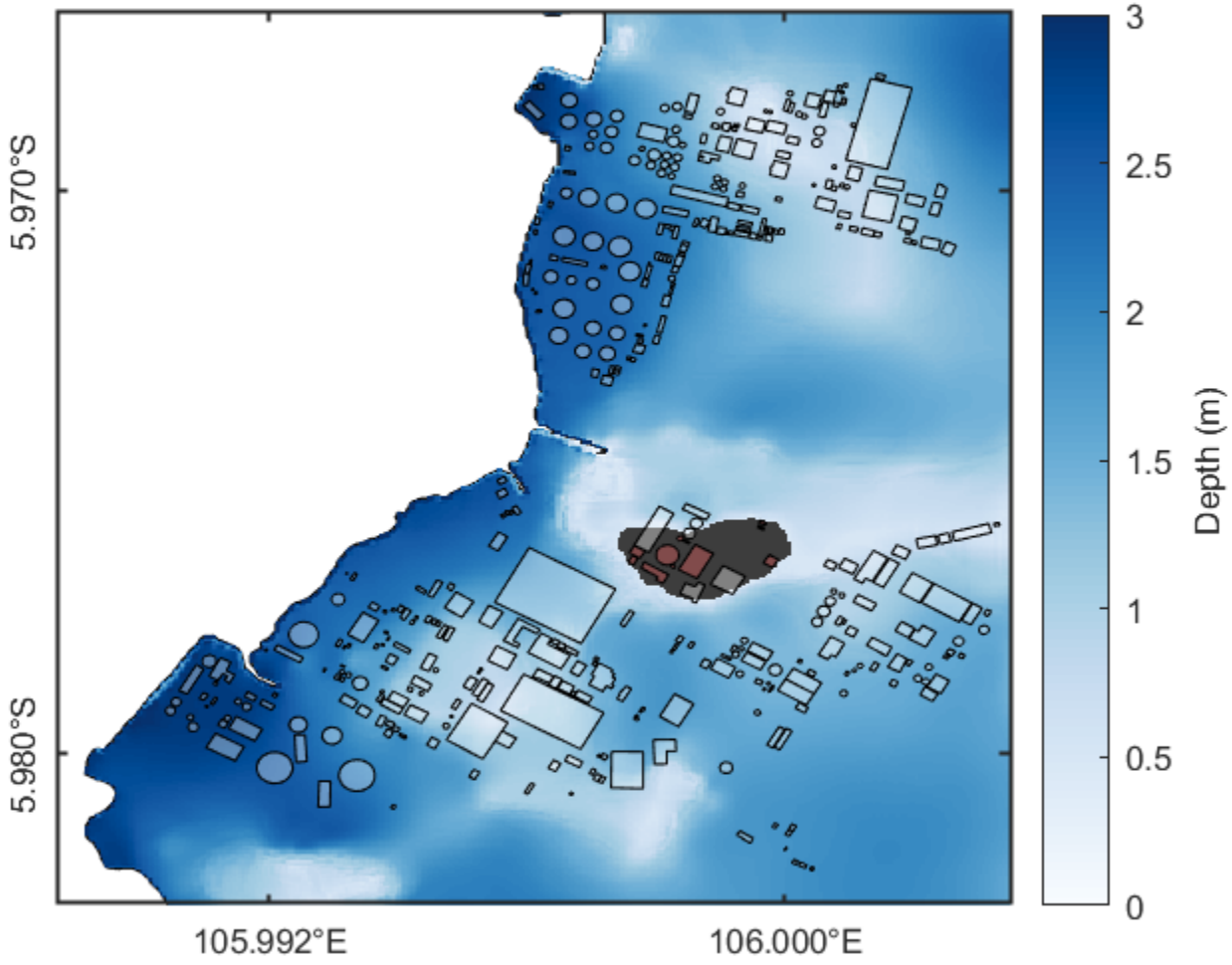


Structure Probability of Exceedance

□ < 0.70 ■ ≥ 0.70

A total of 12 structures have a probability of exceedance ≥ 70% for the selected recovery period.

Structures with a probability of recovery exceeding the threshold within 4.0 to ≤4.5 months

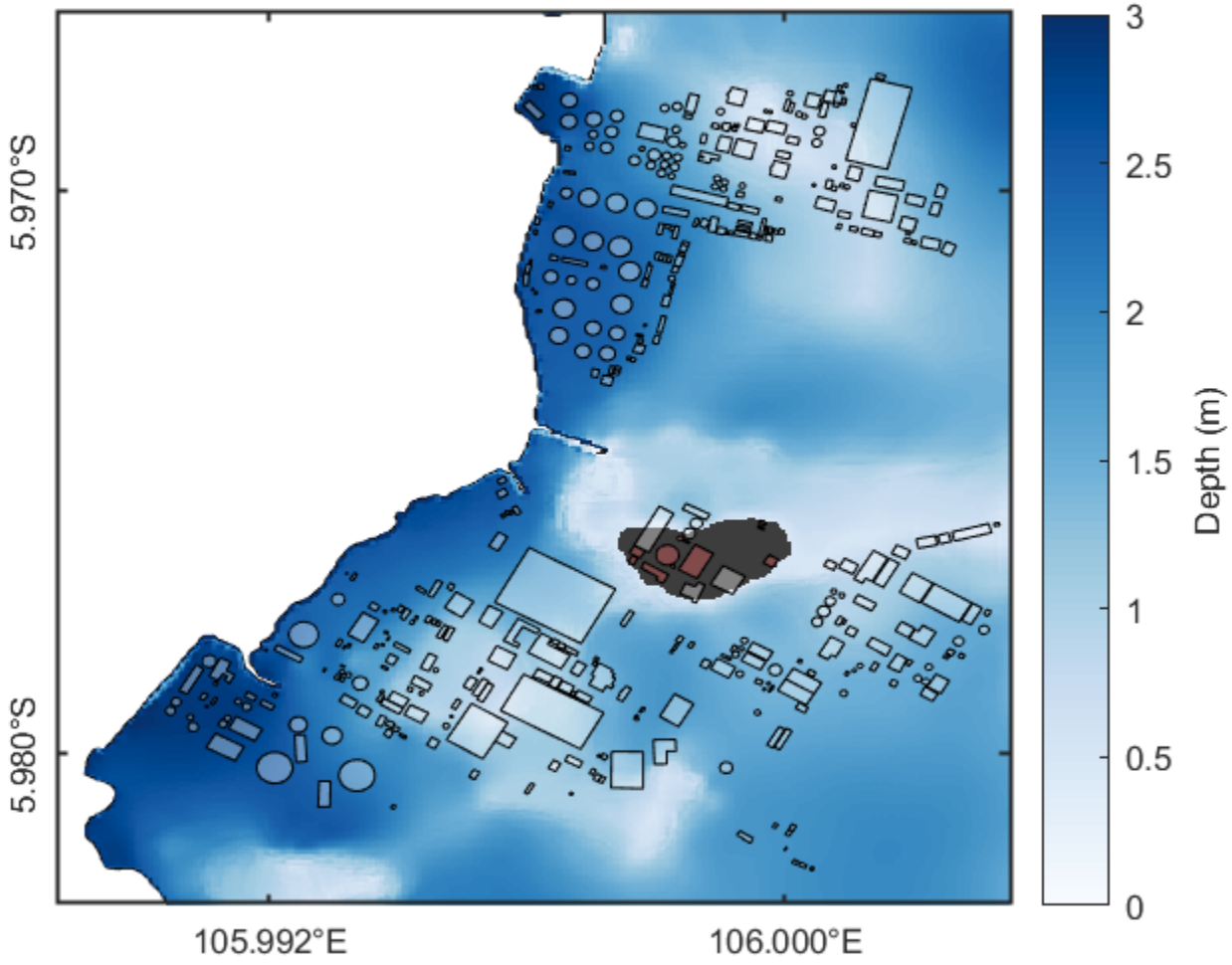


Structure Probability of Exceedance

□ < 0.70 ■ ≥ 0.70

A total of 12 structures have a probability of exceedance ≥ 70% for the selected recovery period.

Structures with a probability of recovery exceeding the threshold within 4.5 to ≤5.0 months

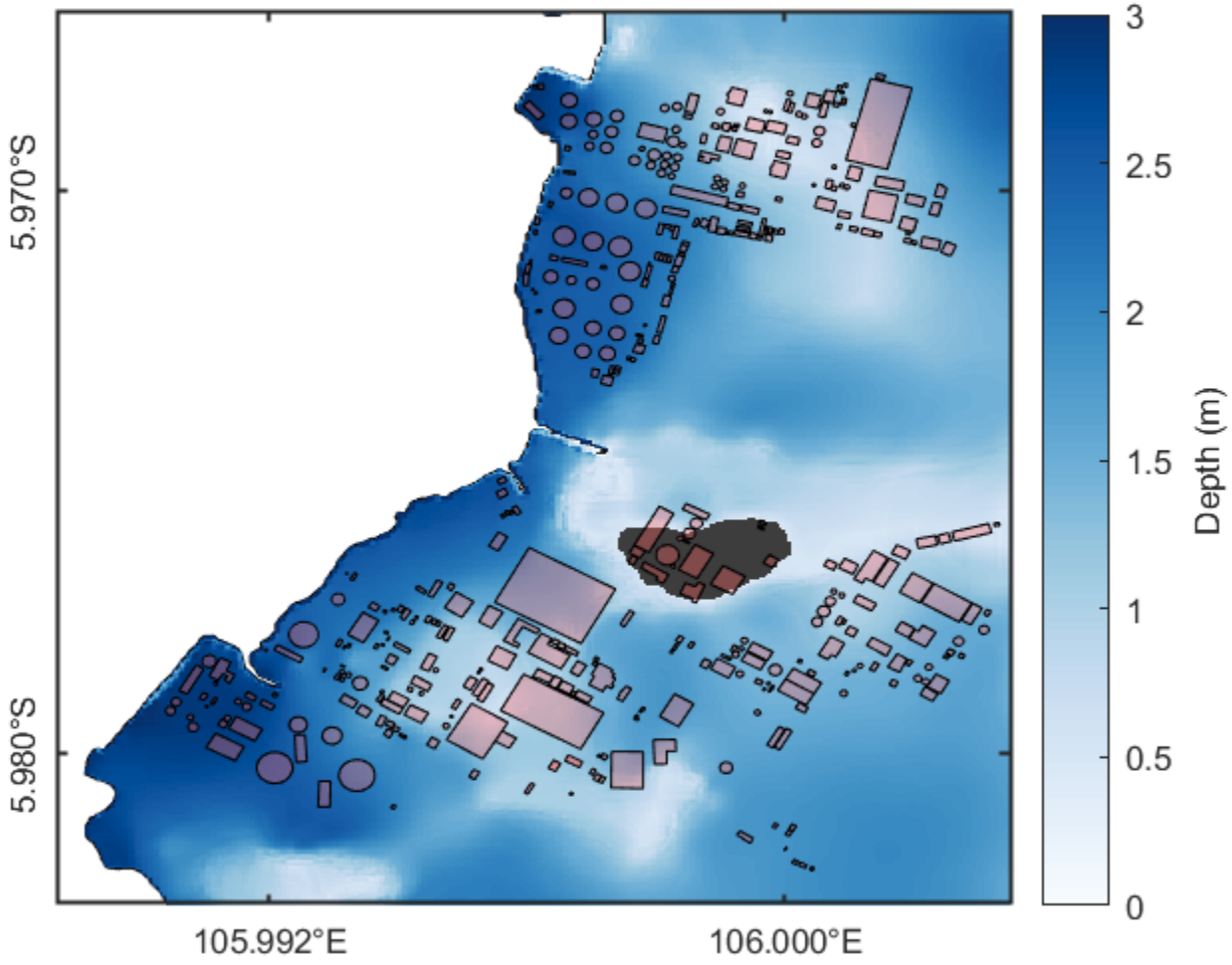


Structure Probability of Exceedance

□ < 0.70 ■ ≥ 0.70

A total of 12 structures have a probability of exceedance ≥ 70% for the selected recovery period.

Structures with a probability of recovery exceeding the threshold within 5.0 to ≤5.5 months

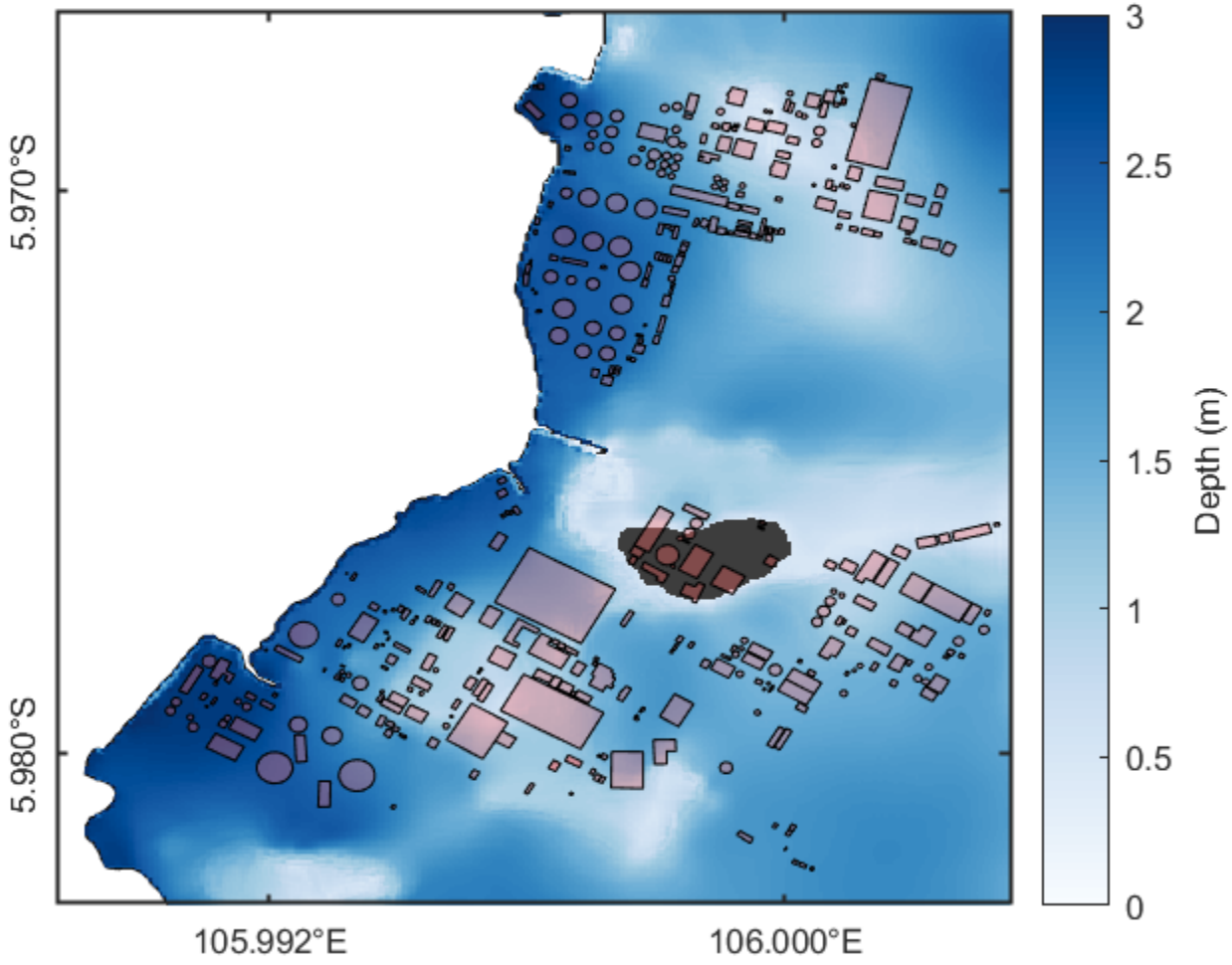


Structure Probability of Exceedance

■ ≥ 0.70 □ < 0.70

A total of 430 structures have a probability of exceedance ≥ 70% for the selected recovery period.

Structures with a probability of recovery exceeding the threshold within 5.5 to ≤6.0 months

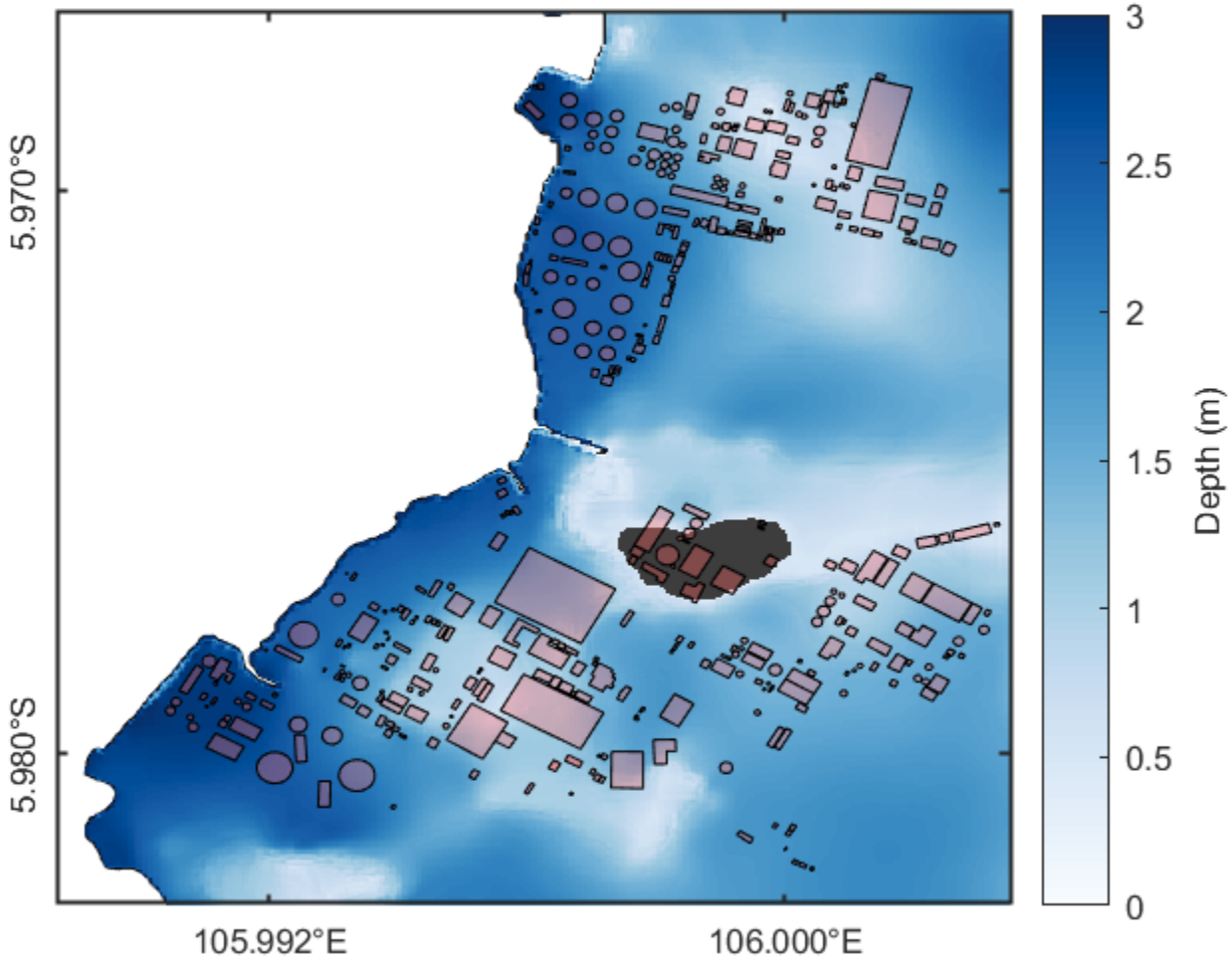


Structure Probability of Exceedance

■ ≥ 0.70 □ < 0.70

A total of 430 structures have a probability of exceedance ≥ 70% for the selected recovery period.

Structures with a probability of recovery exceeding the threshold within 6.0 to ≤6.5 months

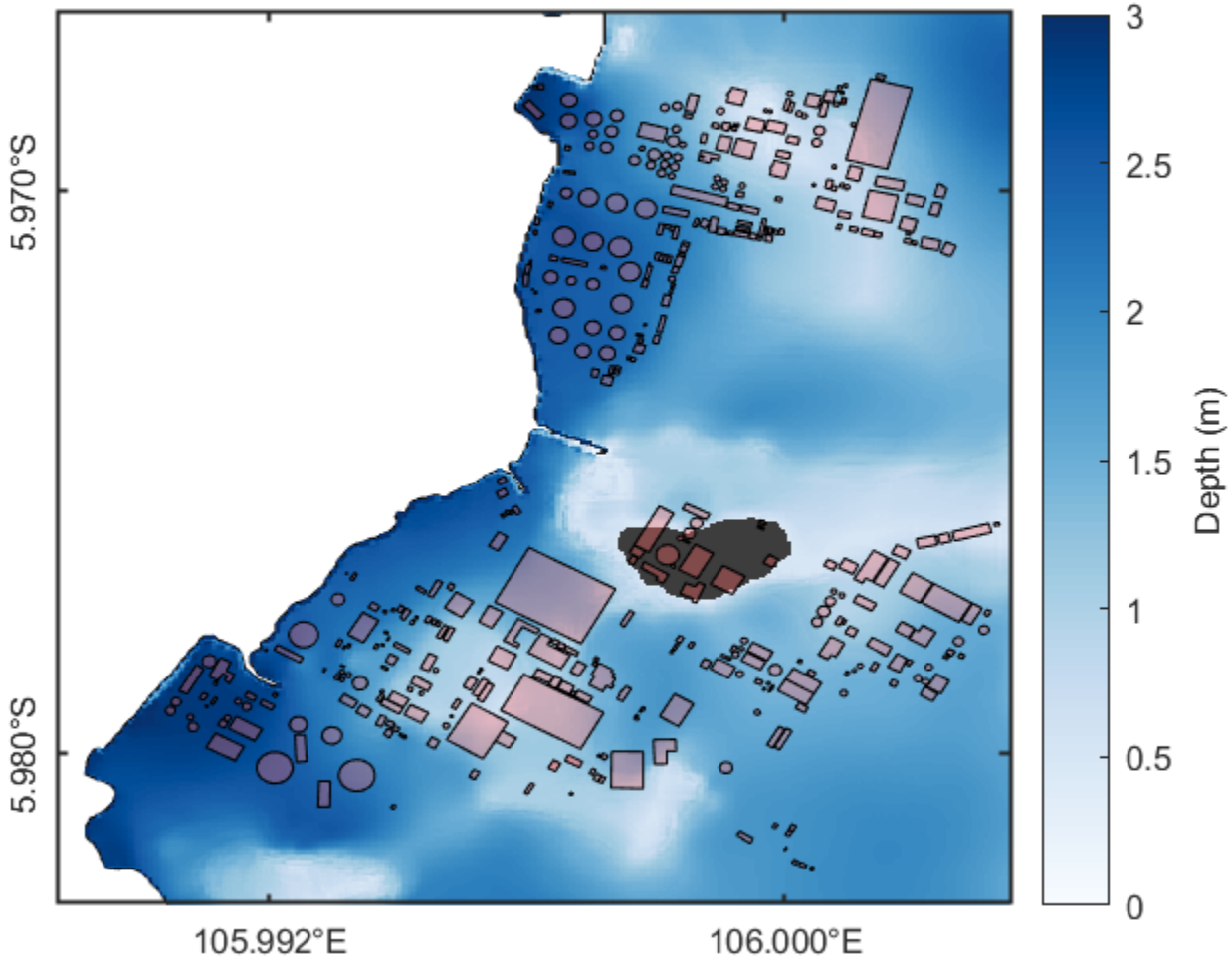


Structure Probability of Exceedance

■ ≥ 0.70 □ < 0.70

A total of 430 structures have a probability of exceedance ≥ 70% for the selected recovery period.

Structures with a probability of recovery exceeding the threshold within 6.5 to ≤7.0 months

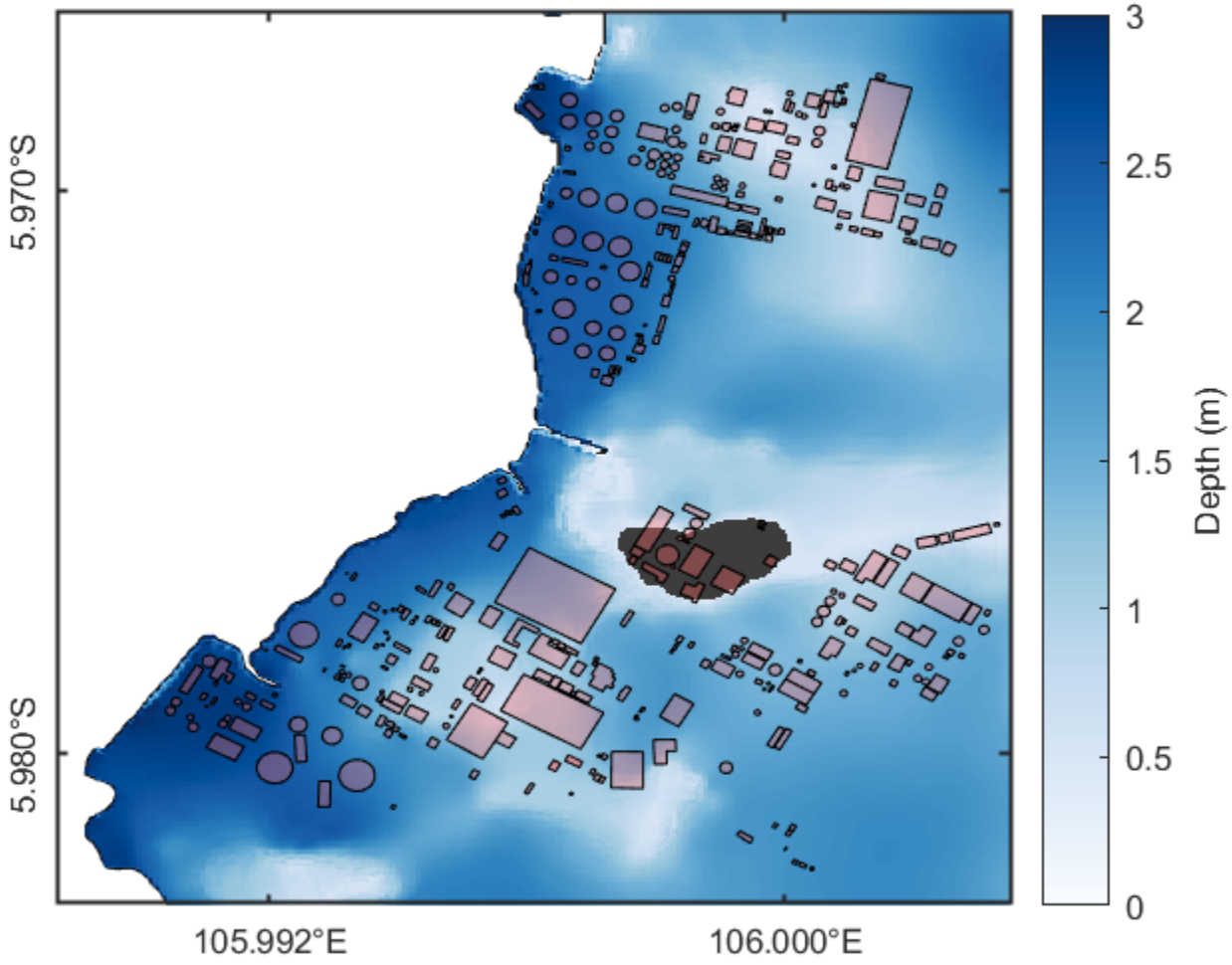


Structure Probability of Exceedance

■ ≥ 0.70 □ < 0.70

A total of 430 structures have a probability of exceedance ≥ 70% for the selected recovery period.

Structures with a probability of recovery exceeding the threshold within 7.0 to ≤7.5 months

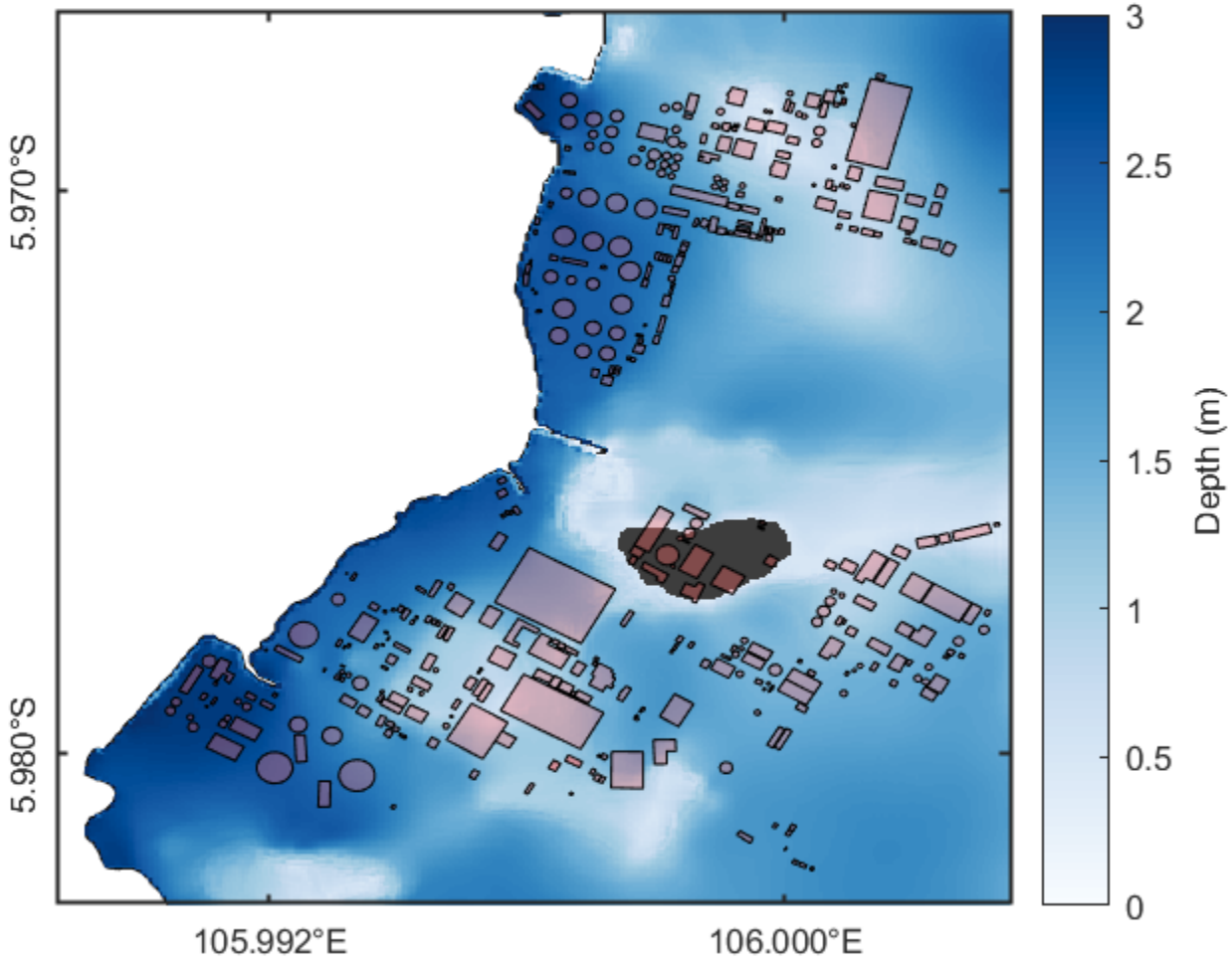


Structure Probability of Exceedance

■ ≥ 0.70 □ < 0.70

A total of 430 structures have a probability of exceedance ≥ 70% for the selected recovery period.

Structures with a probability of recovery exceeding the threshold within 7.5 to ≤8.0 months

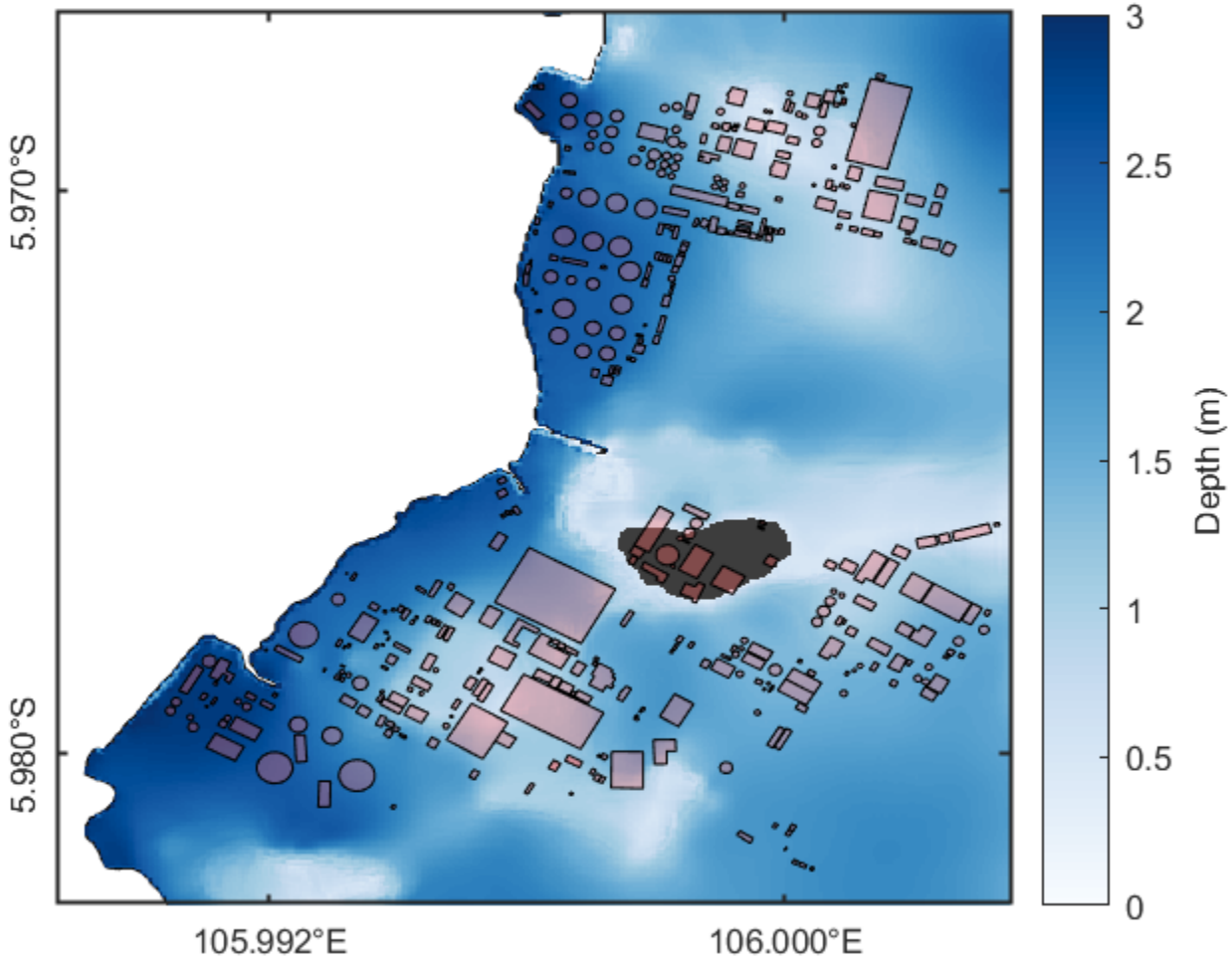


Structure Probability of Exceedance

■ ≥ 0.70 □ < 0.70

A total of 430 structures have a probability of exceedance ≥ 70% for the selected recovery period.

Structures with a probability of recovery exceeding the threshold within 8.0 to ≤8.5 months

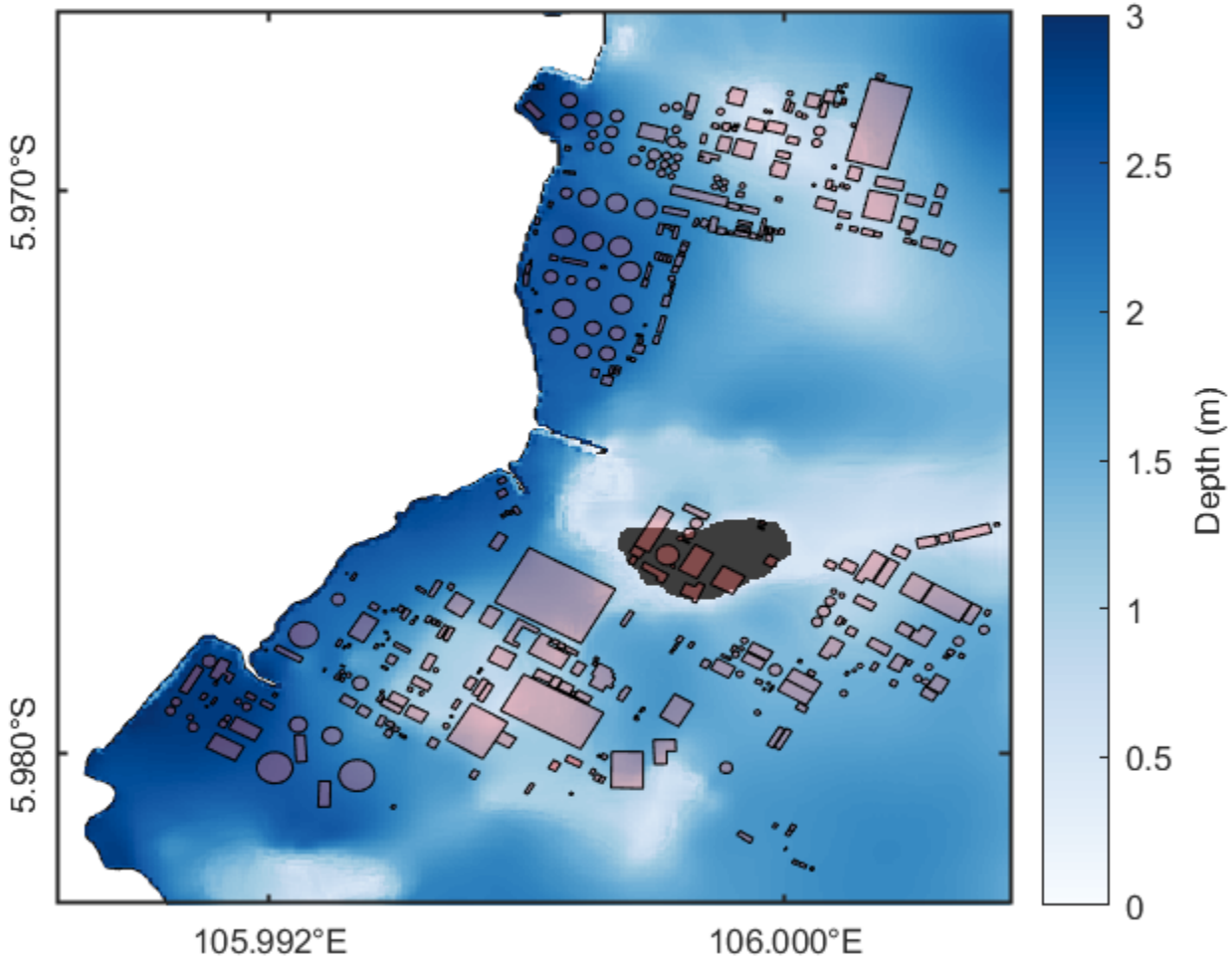


Structure Probability of Exceedance

■ ≥ 0.70 □ < 0.70

A total of 430 structures have a probability of exceedance ≥ 70% for the selected recovery period.

Structures with a probability of recovery exceeding the threshold within 8.5 to ≤9.0 months

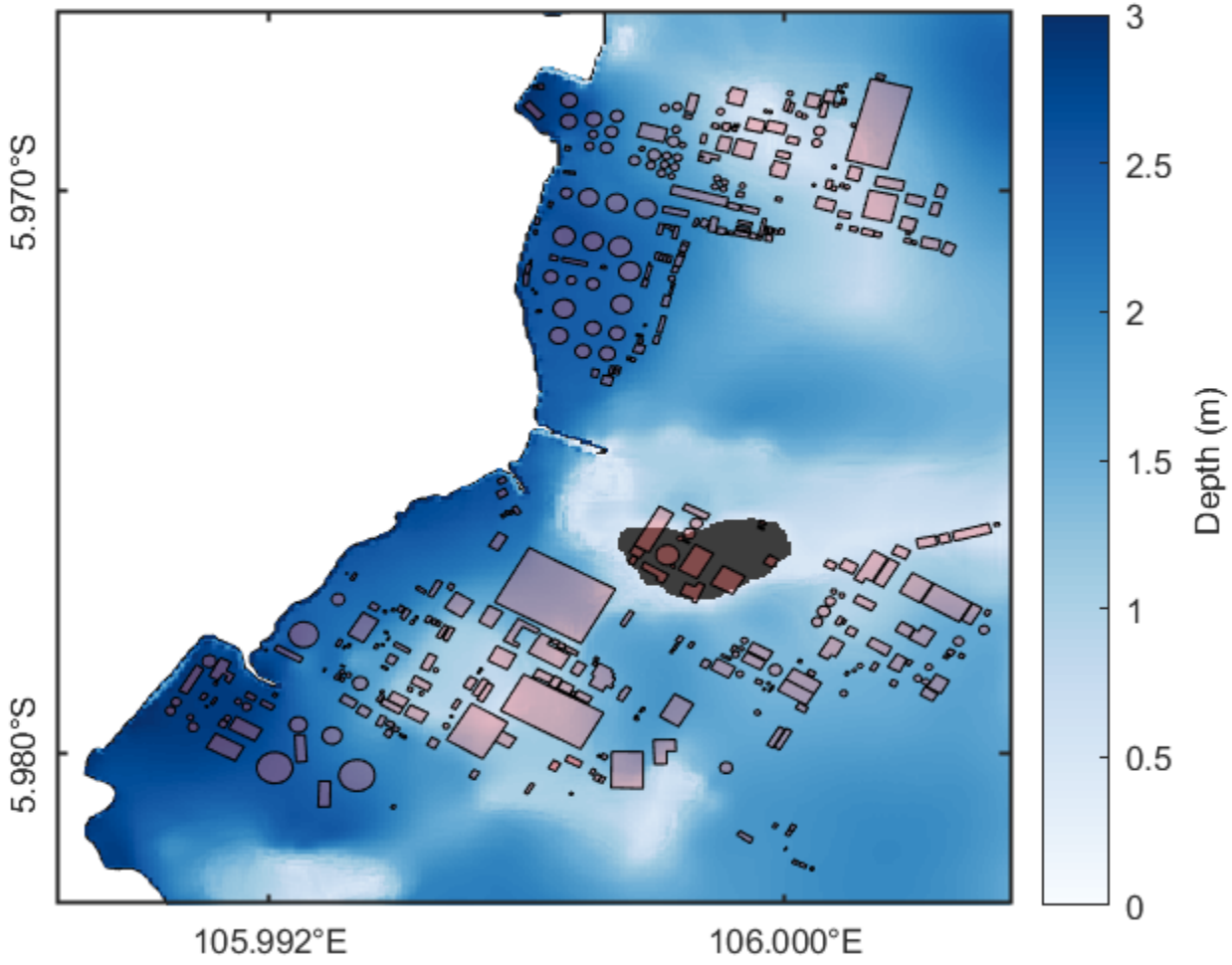


Structure Probability of Exceedance

■ ≥ 0.70 □ < 0.70

A total of 430 structures have a probability of exceedance ≥ 70% for the selected recovery period.

Structures with a probability of recovery exceeding the threshold within 9.0 to ≤9.5 months

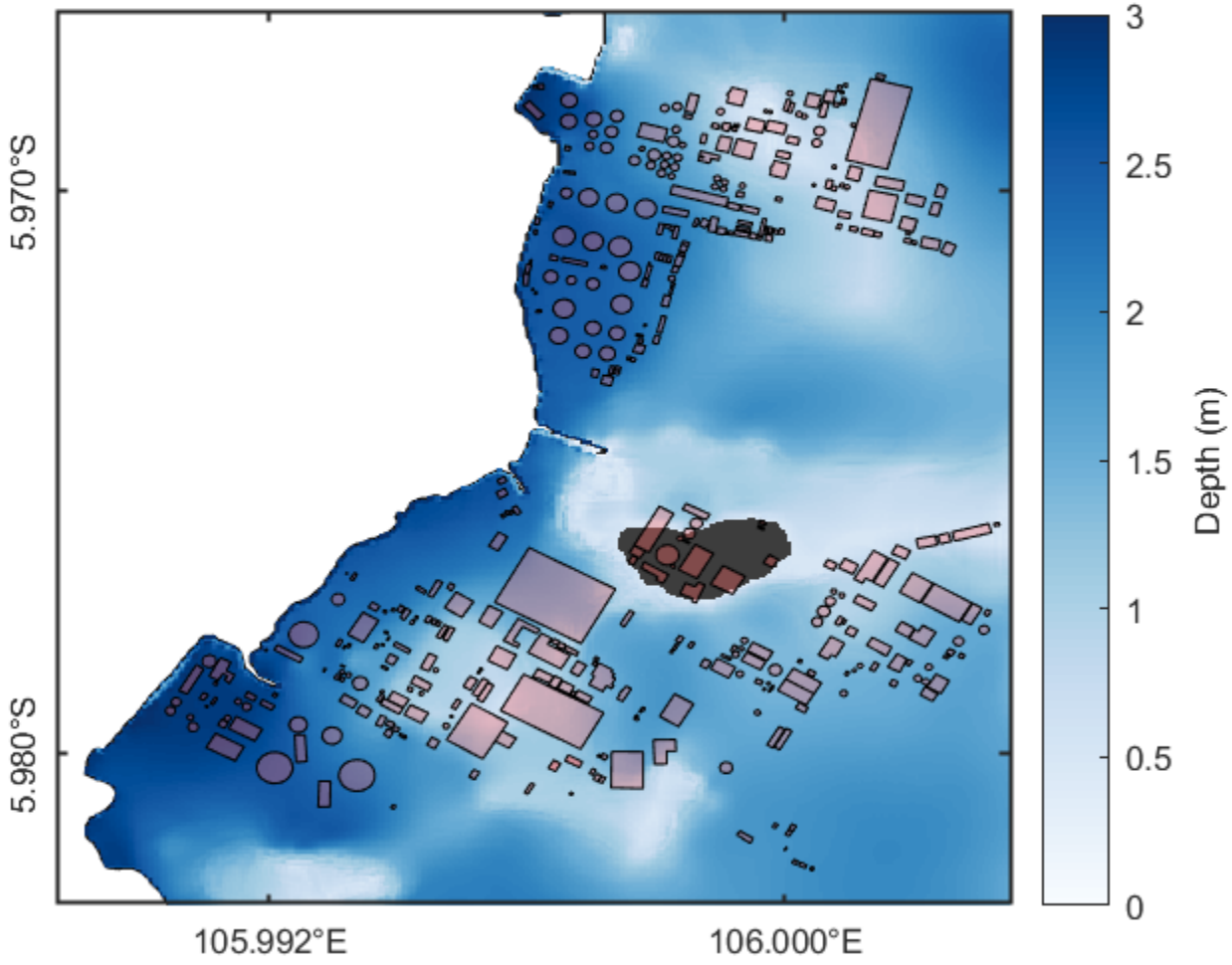


Structure Probability of Exceedance

■ ≥ 0.70 □ < 0.70

A total of 430 structures have a probability of exceedance ≥ 70% for the selected recovery period.

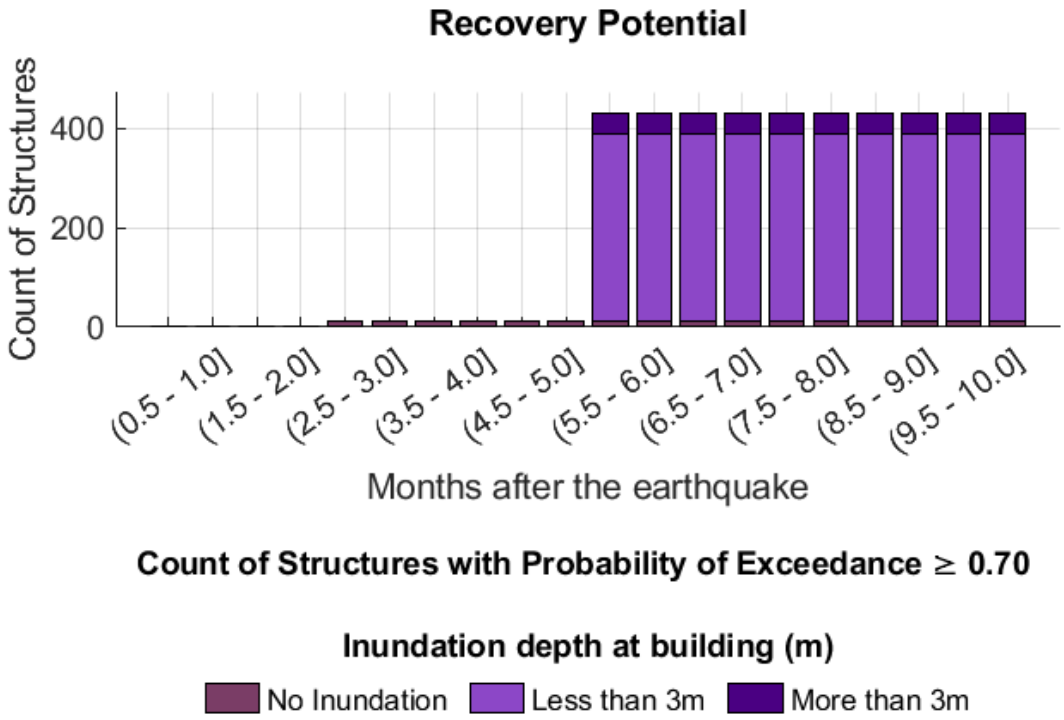
Structures with a probability of recovery exceeding the threshold within 9.5 to ≤10.0 months



Structure Probability of Exceedance

■ ≥ 0.70 □ < 0.70

A total of 430 structures have a probability of exceedance ≥ 70% for the selected recovery period.


Recovery Potential
Count of Structures
400
200
0
(0.5 - 1.0]
(1.5 - 2.0]
(2.5 - 3.0]
(3.5 - 4.0]
(4.5 - 5.0]
(5.5 - 6.0]
(6.5 - 7.0]
(7.5 - 8.0]
(8.5 - 9.0]
(9.5 - 10.0]
Months after the earthquake
Count of Structures with Probability of Exceedance ≥ 0.70
Inundation depth at building (m)
No Inundation
Less than 3m
More than 3m

Months	Total Count	At 0 m	At ≤3m	At >3m
(0.0 - 0.5]	0	0	0	0
(0.5 - 1.0]	0	0	0	0
(1.0 - 1.5]	0	0	0	0
(1.5 - 2.0]	0	0	0	0
(2.0 - 2.5]	12	12	0	0
(2.5 - 3.0]	12	12	0	0
(3.0 - 3.5]	12	12	0	0
(3.5 - 4.0]	12	12	0	0
(4.0 - 4.5]	12	12	0	0
(4.5 - 5.0]	12	12	0	0
(5.0 - 5.5]	430	12	376	42
(5.5 - 6.0]	430	12	376	42
(6.0 - 6.5]	430	12	376	42
(6.5 - 7.0]	430	12	376	42
(7.0 - 7.5]	430	12	376	42
(7.5 - 8.0]	430	12	376	42
(8.0 - 8.5]	430	12	376	42
(8.5 - 9.0]	430	12	376	42
(9.0 - 9.5]	430	12	376	42
(9.5 - 10.0]	430	12	376	42